

```
        ? (ActionResult)new
OkObjectResult($"Hello, {name}")
        : new BadRequestObjectResult("Please
pass a name on the query string or in the
request body");
    }
```

}
In this chapter, you’ve learned how to set up C# on your system, explored its basic syntax, and written a simple C# program. As you continue your coding journey, C# will be an invaluable tool, providing you with the foundation to explore more advanced topics and applications. 

KEY C# LIBRARIES AND FRAMEWORKS			
This table includes descriptions, key features, and typical C# use cases to help you choose the right tools for your projects and understand their applications better.			
Library/ Framework	Description	Key Features	Typical Use Cases
.NET Core/.NET 5+	A cross-platform, high-performance framework for building modern applications.	Cross-platform, high performance, modular, supports web, desktop, mobile, cloud, and IoT applications.	General application development, web services, cloud solutions.
ASP.NET Core	A framework for building web applications and services with .NET Core.	MVC pattern, Razor pages, built-in dependency injection, middleware pipeline.	Web applications, RESTful APIs, microservices.
Entity Framework Core	An object-relational mapper (ORM) for .NET.	LINQ to Entities, code-first and database-first approaches, change tracking.	Data access, database management, CRUD operations.
Xamarin	A framework for building cross-platform mobile applications with C#.	Shared codebase, native performance, access to native APIs.	Mobile applications for iOS, Android, and Windows.
Blazor	A framework for building interactive web UIs with C# instead of JavaScript.	WebAssembly, server-side and client-side hosting models, reusable components.	Single-page applications (SPAs), interactive web applications.
SignalR	A library for adding real-time web functionality to applications.	Persistent connections, connection management, built-in message hubs.	Real-time chat applications, live notifications, gaming.
NUnit	A unit testing framework for .NET languages.	Assertions, test runners, parameterized tests, test fixtures.	Unit testing, automated testing, test-driven development.
AutoMapper	A library for mapping between different object models.	Convention-based mapping, custom value resolvers, nested mappings.	Data transfer objects (DTOs), object transformation.
Dapper	A simple and fast ORM for .NET.	Lightweight, direct SQL execution, supports stored procedures, flexible object mapping.	Data access, performance-critical applications, microservices.
Serilog	A diagnostic logging library for .NET.	Structured logging, sinks for various outputs, enrichers for context.	Logging, application monitoring, debugging.
Hangfire	A library for background job processing.	Persistent storage, job retries, real-time monitoring dashboard.	Background tasks, scheduled jobs, recurring tasks.
MassTransit	A distributed application framework for .NET.	Message-based architecture, support for various message brokers, sagas for long-running processes.	Microservices, distributed systems, event-driven architecture.
Quartz.NET	An enterprise job scheduler for .NET.	Cron-like scheduling, clustering, persistent job storage, transaction support.	Task scheduling, job automation, batch processing.
Polly	A resilience and transient-fault-handling library.	Retry policies, circuit breaker, fallback, bulkhead isolation.	Fault tolerance, resilient applications, API integrations.